



Dissertation on

Explainable Model for Detecting Fetal Brain Anomalies

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1. Introduction

This document outlines the design and implementation of an explainable AI model for detecting fetal brain abnormalities. The primary objective of this project is to develop a robust and interpretable artificial intelligence system capable of accurately identifying various types of fetal brain abnormalities from prenatal ultrasound images. This system aims to assist healthcare professionals in early detection and diagnosis, thereby improving patient outcomes and reducing the risk of complications during pregnancy. Our solution encompasses a multi-stage process, integrating state-of-the-art machine learning algorithms with interpretability techniques to provide transparent and understandable predictions.

2. Current System [if applicable]

In the realm of fetal brain abnormality detection from prenatal ultrasound images, the current diagnostic landscape predominantly relies on the expertise of healthcare professionals, particularly radiologists, who interpret these images based on their clinical experience and domain knowledge. While these professionals demonstrate commendable proficiency in discerning abnormalities, the interpretability and transparency of their decision-making processes often remain opaque, presenting a barrier to understanding and trust among stakeholders.

The existing diagnostic paradigm, although effective to a certain extent, is fraught with inherent limitations, including subjectivity, variability, and the absence of standardized frameworks for decision elucidation. Moreover, the intricacies and nuances underlying the classification of fetal brain abnormalities pose formidable challenges for even the most seasoned practitioners, necessitating a paradigm shift towards more objective, data-driven methodologies.

In response to these challenges, our enhancement initiative endeavors to augment the existing diagnostic framework with an explicit emphasis on explainability and transparency. While the current system excels in leveraging human expertise, it often lacks the ability to articulate the rationale behind diagnostic decisions in a systematic and comprehensible manner. This



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deficiency not only impedes collaborative decision-making but also engenders skepticism and apprehension among stakeholders, thereby necessitating a paradigm shift towards more transparent and interpretable AI-driven solutions.

By enhancing the current system with state-of-the-art explainable AI techniques, we aspire to bridge the gap between empirical data and clinical insights, enabling healthcare professionals to glean actionable insights from model predictions and thereby fostering informed decision-making. Through the judicious integration of interpretability methodologies such as vision transformers, gradcam, we seek to imbue our enhanced system with the capability to elucidate the salient features and patterns underlying each diagnostic decision, thereby engendering trust, understanding, and collaboration among stakeholders.

In essence, our enhancement initiative represents a paradigm shift towards a more transparent, accountable, and collaborative diagnostic framework, one that seamlessly integrates human expertise with the analytical prowess of AI-driven methodologies. Through the fusion of domain knowledge, empirical data, and interpretability, we aspire to usher in a new era of precision and efficacy in the detection and diagnosis of fetal brain abnormalities, thereby advancing the frontiers of prenatal care and enhancing patient outcomes.

3. Design Details

Introduction

This document highlights the High-level design of the project. Our high-level design aims to develop an accurate explainable AI model for the detection of fetal brain anomalies using Vision Transformer and LRP.

Current Systems

Many existing systems focus only on fetal brain anomaly detection and focus primarily only on achieving high accuracy in classification task. The existing systems lack transparency and interpretability in their decision-making process, which we would be implementing through our project.

Design Goals

The goals include

- Develop an accurate and reliable model for detection of fetal brain anomalies.
- Fine tune the Vision Transformer parameters to achieve better accuracy in classification .
- Enable early detection of fetal brain anomalies to facilitate timely interventions and medical interventions during gestation period.
- Implement LRP explainability to provide clear and reasoning for the decision made by ViT model.
- Highlight the important features or regions within fetal brain images that are based on which the classification is made by the model.

Architecture Choices

- Vision Transformer (ViT) for image classification: Processes images as sequences of tokens or patches, allowing it to capture global information.
- Layer-wise Relevance Propagation (LRP) for explainability: LRP is a state-of-the-art technique that generates heat maps highlighting important regions in input images.

Constraints, Assumptions and Dependencies

Constraints:

- Availability of computing resources for model training and inference.
- Unavailability of Large Dataset may limit the model's training efficacy and generalization capabilities.
- Unbalanced dataset can introduce bias into the model's predictions.
- Validation of the model's predictions through medical expert intervention may be challenging

Assumptions:

- We assume that the annotations provided by medical experts for the fetal brain images are consistent and accurately reflect the presence or absence of anomalies.
- We assume that the training dataset is representative of the population distribution of fetal brain images encountered in clinical practice.
- We assume that the model developed will generalize well to unseen fetal brain images from different sources or populations.

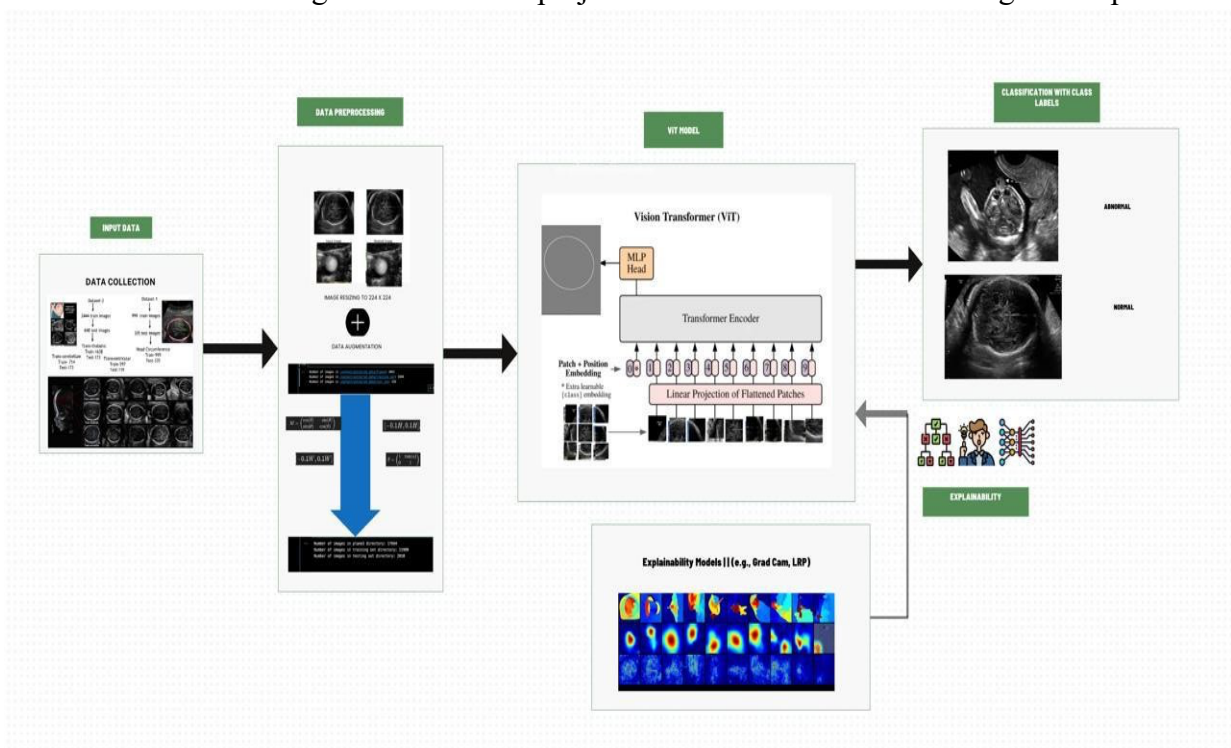
Dependencies

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- Availability of labeled fetal brain image datasets for training and evaluation.
- Cloud service providers for deployment and management of the model on cloud-based infrastructure.
- Collaboration with medical professionals and domain experts for dataset annotation, model validation, and clinical integration.

Low Level Design

The low-level design of this project contains the following components



Data Collection

Dataset 1: Contains images in different planes of the fetus collected during the gestational period; only brain planes (trans-thalamic, trans-cerebellar, and transventricular) were extracted from 12,000 images. This dataset comprises 2949 images.

Dataset 2: Consists of images of fetal brain head circumference with 1998 training images and 335 additional images.

Both the datasets were annotated by medical experts.

Data Preprocessing

All images are standardized to a resolution of 224×224 pixels.

Augmentation techniques are applied to enhance the diversity of the training data, which helps prevent overfitting and improve model generalization.

ViT (Vision Transformer)

Explainability Model

The LRP will generate heatmaps that highlight important regions in the input images, indicating which parts contribute most significantly to the model's classification decision.

Design Description

For data collection, we gather two distinct datasets. Dataset 1 comprises 2949 images of fetal brain planes, including trans-thalamic, trans-cerebellar, and transventricular views, extracted from a larger collection of 12,000 fetal images. Dataset 2 consists of 1998 training images and 335 additional images depicting fetal brain head circumference. Both datasets are annotated by medical experts. Following data collection, images undergo preprocessing, standardized to 224×224 pixels resolution,

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and augmented to diversify training data, reducing overfitting risks and enhancing model generalization. Subsequently, preprocessed images are fed into a Vision Transformer (ViT) for feature learning and classification, distinguishing normal from abnormal fetal brain images. For model explainability, we employ Layer-wise Relevance Propagation (LRP), generating heat maps highlighting crucial image regions influencing classification decisions. Utilizing GPUs available on platforms like Google Colab or Jupyter Notebooks accelerates model training, while cloud services streamline the overall process, facilitating data storage, model deployment, and computational scalability.

Reusability Conditions

The datasets collected for fetal brain anomaly detection can be reused for future researchs and model development.

Preprocessing techniques and augmentation strategies employed on the image datasets can be reused for similar medical imaging tasks.

The Vision Transformer (ViT) architecture and Layer-wise Relevance Propagation (LRP) explainability model can be reused or adapted for other image classification tasks beyond fetal brain anomaly detection.

External Interfaces

Collaboration with Radiologists and Medical domain experts to provide feedback on model decision making.

It is possible to integrate with external medical imaging databases or repositories in the future to provide access to additional fetal brain image data.

Design Details

Fetal brain images are resized to a standardized resolution of 224×224 pixels to ensure consistency in input dimensions across the dataset. Image pixel values are normalized to a common scale (e.g., $[0, 1]$ or $[-1, 1]$) to facilitate convergence during model training.

The ViT architecture consists of multiple layers of self-attention mechanisms and feed-forward neural networks. The number of layers, attention heads, and hidden units are configurable parameters.

At the final layer output, LRP is implemented. From the output layer to the input layer of the neural network, relevance scores are propagated backward, highlighting important regions of the input image that contribute to the model's prediction. By visualizing the relevance scores assigned to each pixel in the input image, LRP generates heat maps.

TensorFlow or PyTorch, which support GPU computing, are recommended libraries for training and implementing deep learning operations. It may be possible to improve the performance of mixed precision models by optimizing batching, parallelizing, and training them.

Cloud-based services such as Google Colab and AWS SageMaker are used to train models as they provide scalable compute resources and pre-configured deep learning environments.

3.1 Novelty

- Introduction of Vision Transformer (ViT) architecture for fetal brain anomaly detection in ultrasound images, which represents a novel approach in medical imaging.
- Integration of state-of-the-art explainable AI techniques with ViT to enhance interpretability and transparency in deep learning models for medical diagnostics.

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3.2 Innovativeness

- Utilization of cutting-edge deep learning and XAI methods to address the opacity and interpretability challenges in medical image analysis, specifically in fetal medicine.
- Adoption of a multidisciplinary approach that combines expertise in artificial intelligence, medical imaging, and clinical diagnostics to develop innovative solutions for prenatal care.

3.3 Interoperability

- Ensuring compatibility with existing healthcare systems and standards for seamless integration into clinical workflows.
- Designing the model to accept input from various ultrasound imaging devices and formats commonly used in clinical practice, enhancing interoperability across different healthcare settings.

3.4 Performance

- Optimization of model performance through hyperparameter tuning, regularization techniques, and model compression for efficient inference.
- Validation of model performance against standard classification metrics and benchmarks to ensure high accuracy and reliability in fetal brain anomaly detection.

3.5 Security

- Implementation of data encryption, access controls, and secure communication protocols to protect patient privacy and sensitive medical information.
- Adherence to regulatory requirements such as HIPAA (Health Insurance Portability and Accountability Act) to ensure compliance with healthcare data security standards.

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3.6 Reliability

- scenarios to ensure robustness and stability.
- Incorporation of error-handling mechanisms and quality assurance processes to mitigate risks and ensure reliable performance in real-world applications.

3.7 Maintainability

- Documentation of codebase, model architecture, and implementation details to facilitate ease of maintenance and future updates.
- Adoption of modular and scalable design principles to enable seamless integration of new features, enhancements, and bug fixes.

3.8 Portability

- Containerization of the model using Docker to ensure portability across different computing environments and platforms.
- Designing the model with platform-agnostic dependencies to enhance flexibility and ease of deployment in diverse healthcare settings.

3.9 Legacy to modernization

- Bridging the gap between traditional medical imaging diagnostics and AI-enabled solutions by leveraging state-of-the-art deep learning and XAI techniques.
- Facilitating the transition from legacy diagnostic approaches to modern AI-driven methodologies in prenatal care, thereby improving clinical outcomes and patient care.

3.10 Reusability

- Designing modular and reusable components to facilitate the reuse of the model architecture and XAI techniques in other medical imaging applications.
- Sharing best practices, codebase, and methodologies with the research community to promote collaboration and accelerate advancements in AI-driven healthcare solutions.

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3.11 Application compatibility

- Ensuring compatibility with various ultrasound imaging devices and formats commonly used in clinical practice to maximize applicability and adoption.
- Designing user-friendly interfaces and tools for clinicians to interact with the model, interpret results, and integrate them into patient care decisions seamlessly.

3.12 Resource utilization, Etc.,

- Optimizing resource utilization by leveraging cloud computing resources for model training and inference, ensuring scalability and cost-effectiveness.
- Implementing efficient data processing and storage techniques to minimize computational overhead and maximize resource efficiency in model deployment.

Appendix A: Definitions, Acronyms and Abbreviations

Definitions

Explainable AI (XAI): Artificial Intelligence methods and techniques that make the results of the AI model understandable to humans.

Fetal Brain Abnormalities: Irregularities in the development of a fetus's brain that can be detected through prenatal imaging.

Prenatal Ultrasound Images: Images captured during an ultrasound scan of a pregnant woman, used to examine the fetus.

Vision Transformer (ViT): A neural network model for image classification that treats an image as a sequence of patches.

Layer-wise Relevance Propagation (LRP): A technique for explaining predictions of machine learning models by decomposing the output and assigning relevance scores to the input features.

Acronyms and Abbreviations

AI: Artificial Intelligence

LRP: Layer-wise Relevance Propagation



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ML: Machine Learning

ViT: Vision Transformer

XAI: Explainable Artificial Intelligence

GPU: Graphics Processing Unit

HIPAA: Health Insurance Portability and Accountability Act

AWS: Amazon Web Services

Additional Terms

Grad-CAM: Gradient-weighted Class Activation Mapping, a technique for visualizing important regions in an image for classification tasks.

Self-Attention Mechanisms: Components of a neural network that allow it to focus on different parts of the input sequence for better understanding and prediction.

Data Augmentation: Techniques used to increase the diversity of a training dataset by applying random transformations.

Model Generalization: The ability of a machine learning model to perform well on new, unseen data.

Model Validation: The process of evaluating the performance of a machine learning model on a separate dataset not used during training.

Appendix B: References

[1]X. P. Burgos-Artizzu et al., “Evaluation of deep convolutional neural networks for automatic classification of common maternal fetal ultrasound planes,” Scientific Reports, vol. 10, no. 1, Jun. 2020, doi: <https://doi.org/10.1038/s41598-020-67076-5>.

[2]S. Stassin, V. Corduant, S. A. Mahmoudi, and X. Siebert, “Explainability and Evaluation of Vision Transformers: An In-Depth Experimental Study,” Electronics, vol. 13, no. 1, p. 175, Jan. 2024, doi: <https://doi.org/10.3390/electronics13010175>.

[3]H. N. Xie et al., “Using deep-learning algorithms to classify fetal brain ultrasound images as normal or abnormal,” Ultrasound in Obstetrics & Gynecology, vol. 56, no. 4, pp. 579–587, Oct. 2020, doi: <https://doi.org/10.1002/uog.21967>.

[4]S. Shivaprasad, S. Shanbhog and J. Medikonda, "Detecting Fetal Brain Abnormalities Using Deep Learning Technique," 2023 International Conference on Recent Advances in Information Technology for Sustainable Development (ICRAIS), Manipal, India, 2023, pp. 234-238, doi: 10.1109/ICRAIS59684.2023.10367109.

keywords: {Deep learning;Learning systems;Ultrasonic imaging;Transfer learning;Brain modeling;Convolutional neural networks;Prognostics and health management;Fetal brain abnormalities;Deep learning;Convolution neural network;Transfer learning;Ultrasound},

[5]P. Komorowski, H. Baniecki and P. Biecek, "Towards Evaluating Explanations of Vision Transformers for Medical Imaging," 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Vancouver, BC, Canada, 2023, pp. 3726-3732, doi: 10.1109/CVPRW59228.2023.00383. keywords:

{Visualization;Sensitivity;Magnetic resonance imaging;Predictive models;Transformers;Pattern recognition;Reliability},

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Agnostic Framework for Explaining Black Box Models," 2023 IEEE International Conference on Data Mining (ICDM), Shanghai, China, 2023, pp. 944-949, doi: 10.1109/ICDM58522.2023.00105.

keywords: {Training;Measurement;Closed box;Predictive models;Transformers;Natural language processing;Optimization;Explainable AI;computer vision;transformers},

[7]Baek, Ji-Won & Chung, Kyungyong. (2023). Explainable Anomaly Detection Using Vision Transformer Based SVDD. Computers, Materials & Continua. 74. 6573-6586. 10.32604/cmc.2023.035246.

[8][1]Y. Lei, X. Wang, J. Fang, Yi-Ping Phoebe Chen, and L. Chi, "eX-ViT: A Novel explainable vision transformer for weakly supervised semantic segmentation," Pattern Recognition, vol. 142, pp. 109666–109666, Oct. 2023, doi: <https://doi.org/10.1016/j.patcog.2023.109666>.

[9]K. Nagabhushan, S. Hegde, S. F. Sheikh, S. J. Bharadwaj and V. R. B. Prasad, "Deep Learning Approaches for Early Diagnosis and Prediction of Fetal Abnormalities," 2023 7th International Conference on Imaging, Signal Processing and Communications (ICISPC), Kumamoto, Japan, 2023, pp. 54-59, doi: 10.1109/ICISPC59567.2023.00018. keywords: {Ultrasonic imaging;Ultrasonic variables measurement;Nose;Medical services;Predictive models;Signal processing;Transformers;Fetal healthcare;fetal abnormality;prenatal healthcare;early detection;ultrasound images;nuchal translucency;cystic hygroma;absent nasal bone;ventriculomegaly;vision transformer;convolutional neural networks;machine learning},

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Appendix C: Record of Change History

#	Date	Document Version No.	Change Description	Reason for Change
1.	1/2/24	1	Changed from Deep learning models to ViTs	To improve accuracy
2.	25/3/24	2	Use of LRP for explainability	Best for ViTs(found in research paper)
3.	5/4/24	3	Included extra images for dataset	To improve performance of the model

Appendix D: Traceability Matrix

Project Requirement Specification Reference Section No. and Name.	DESIGN / HLD Reference Section No. and Name.
1.1 Introduction	Introduction
1.2 Project Scope	Current System
2.1 Data Collection	3.1 Novelty
2.2 Model Construction	3.2 Innovativeness
2.3 Integration of Explainability	3.3 Interoperability
2.4 Evaluation and Verification	3.4 Performance
3.1 Research Objectives	3.5 Security
4.1 Operating Environment	3.6 Reliability
4.2 Constraints, Assumptions and Dependencies	3.7 Maintainability
4.3 Risks and Obstacles	3.8 Portability
4.4 Functional Requirements	3.9 Legacy to Modernization
4.5 Non-Functional Requirements	3.10 Reusability



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4.5.1 Performance Requirements	3.11 Application Compatibility
4.5.2 Safety Requirements	3.12 Resource Utilization